INTRODUCTION TO FLUID MECHANICS

Department of Chemical Engineering

CHEM ENG 2O04
Course Notes

THE BERNOULLI EQUATION – Part II

Fox & McDonald
10th & 9th editions: 8.6 – 8.8

PIPE-FLO
Tutorial #5 'Fluid Statics with Minor Losses'
Tutorial #6 'Other devices, valves, & meters'

“Investing in yourself is the best thing you can do. Anything that improves your own talents; nobody can tax it or take it away from you. They can run up huge deficits and the dollar can become worth far less. You can have all kinds of things happen. But if you’ve got talent yourself, and you’ve maximized your talent, you’ve got a tremendous asset that can return ten-fold.”

Warren Buffet

Turbulent flow in non-circular ducts

- The Bernoulli equation can also be used for calculations involving non-circular ducts
  - Square/rectangular are OK if the cross sections are not too "exaggerated"
  - Elliptical ducts
  - Various other shapes

OK for $\frac{1}{4} < ar < 4$, where

$$\text{aspect ratio (ar)} = \frac{\text{height (h)}}{\text{width (b)}}$$

- Substitute 'hydraulic diameter (D_h)' for pipe diameter:

Eqn 8.50 $$D_h \equiv \frac{4A}{P}$$

Cross-sectional area

'Wetted' perimeter (the portion of the duct perimeter that is physically touching the fluid)

Turbulent flow in non-circular ducts

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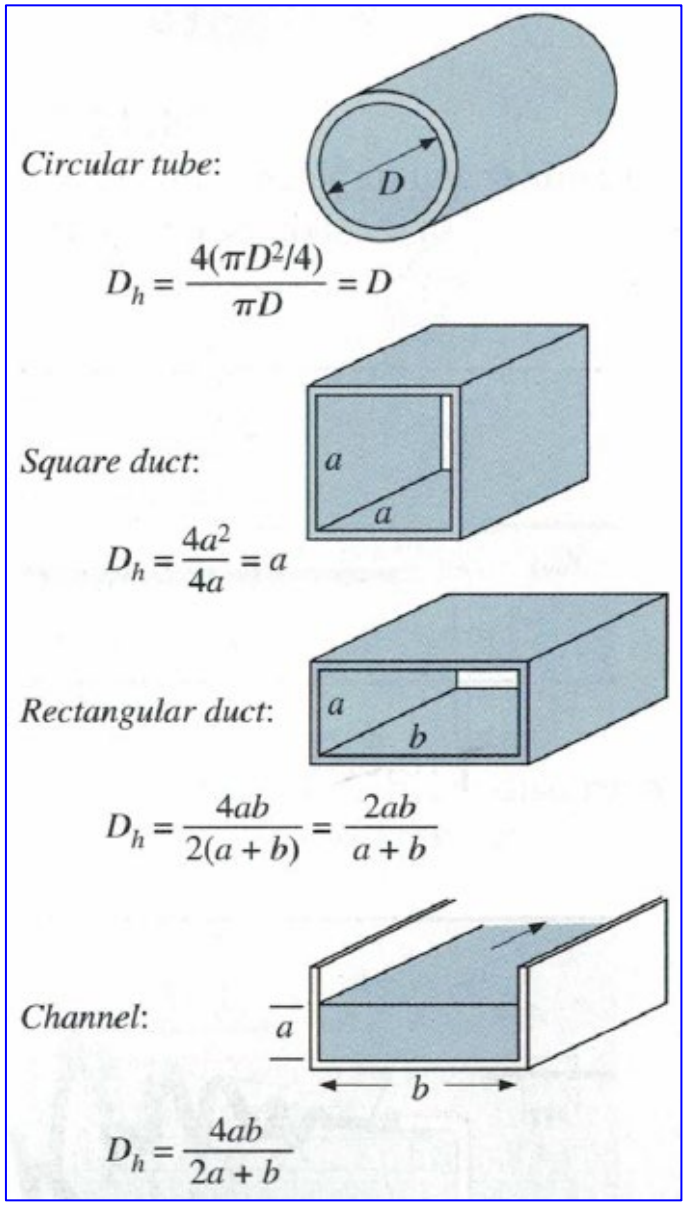


Turbulent flow in non-circular ducts

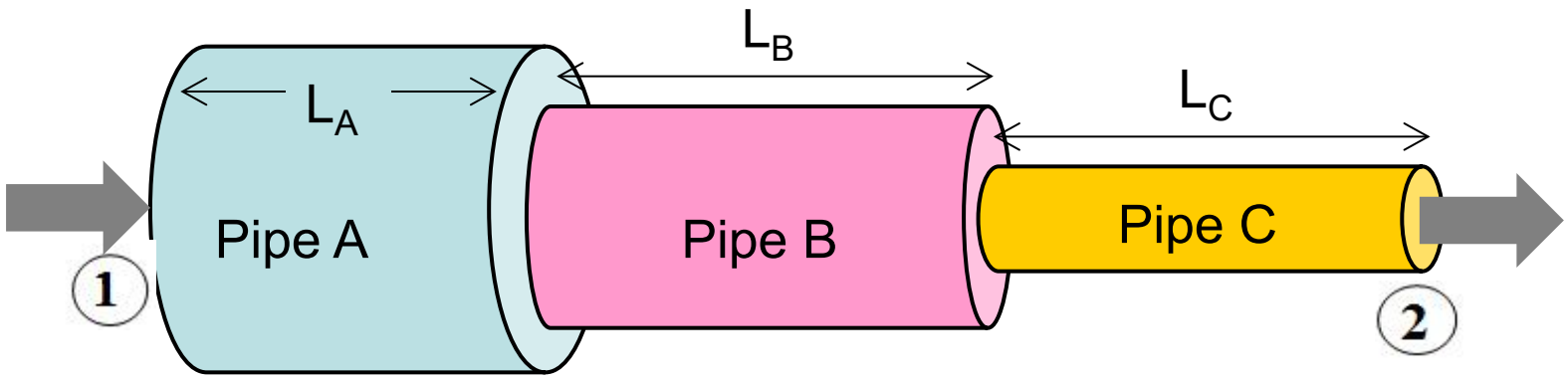
- Reynolds number: $$\mathrm{Re}_{D_h} = \frac{\rho V D_h}{\mu}$$

- Friction factor: $$f_{D_h} = \frac{0.316}{\mathrm{Re}_{D_h}^{0.25}}$$

$$f_{D_h} = \left[\frac{1}{-1.8\log_{10}\left[\left(\frac{\varepsilon}{3.7}\right)^{1.11} + \frac{6.9}{\mathrm{Re}_{D_h}}\right]}\right]^2$$

  - NOTE – this is only an approximation
    ~15% uncertainty in f for turbulent flow

Frictional losses are cumulative



h_{lT} accounts for total energy loss

Need to account for:

1. Irreversible conversion of mechanical energy at Location 1 to unwanted thermal energy
2. Loss of energy via heat transfer

$$h_{l_T} = h_l + h_{l_m}$$

Dependent on flow profile through the duct

Independent of pipe orientation

Major losses due to frictional effects

Minor losses due to entrances, fittings (elbows, valves), area changes, etc.

Bernoulli Equation – Minor losses (h_{lm})

Minor losses due to entrances, fittings (elbows, valves), area changes, etc.

$$h_{lm} = \sum \left[K \frac{V_c^2}{2} \right]$$

Eqn 8.40

Loss coefficient (K)

Characteristic velocity (V_c)

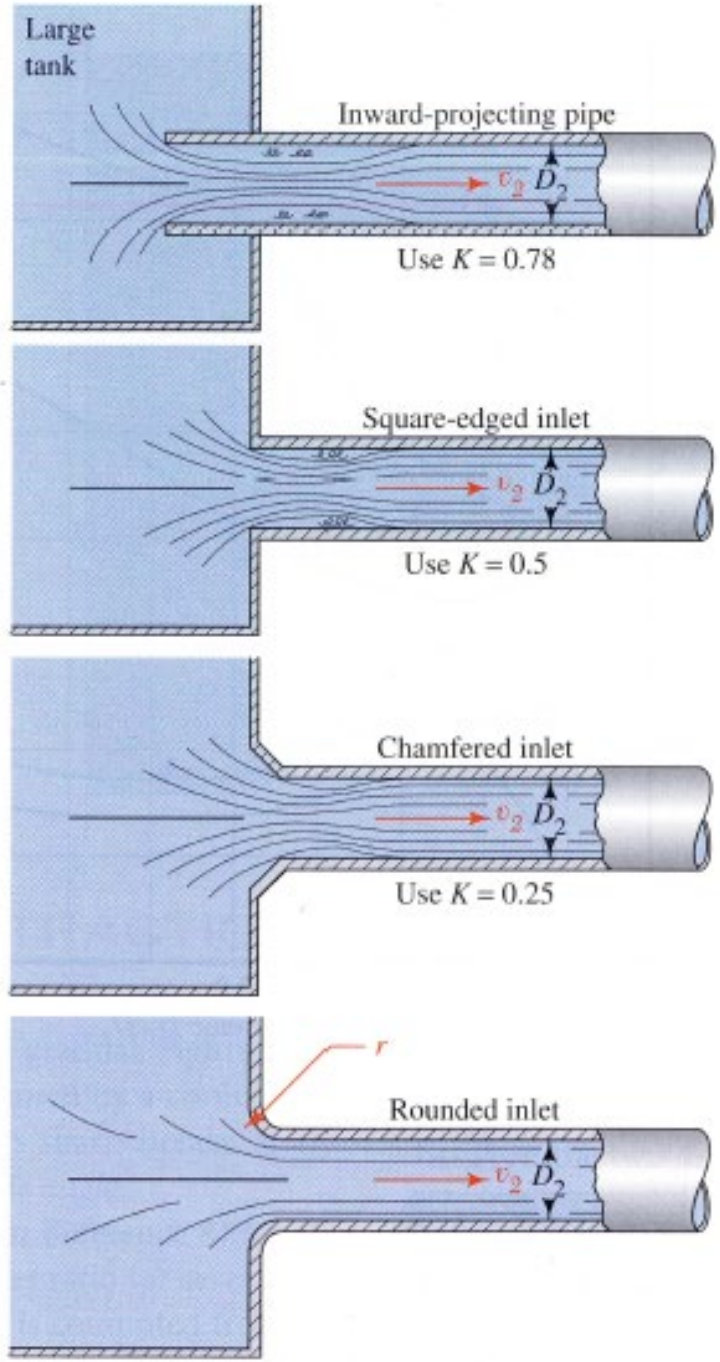
Look these up in various tables & figures!

FIGURE 10.14 Entrance resistance coefficients.

Mott & Untener Applied Fluid Mechanics 7th edition

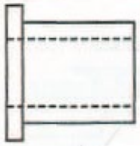
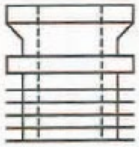
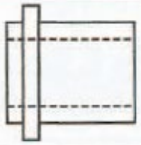
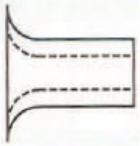
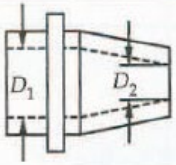
$$h_{lm} = K_{entrance}\frac{\bar{V}^2}{2}$$

where $\bar{V}$ is the velocity in the pipe



r/D_2	K
0	0.50
0.02	0.28
0.04	0.24
0.06	0.15
0.10	0.09
>0.15	0.04 (Well-rounded)

Tables of Loss Coefficients - Janna textbook (3rd edition)

 Square edged inlet $K = 0.5$	 Basket strainer $K = 1.3$
 Re-entrant inlet or inward projecting pipe $K = 1.0$	 Well rounded inlet or a bell mouth inlet $K = 0.05$
Foot valve $K = 0.8$	Exit $K = 1.0$
 Convergent outlet or nozzle $K = 0.1(1 - D_2/D_1)$ D_2/D_1 from 0.5 to 0.9	

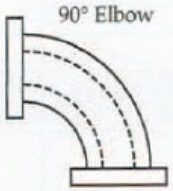
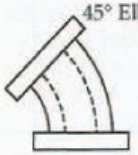
	threaded	flanged, welded, glued, bell & spigot
 90° Elbow	regular $K = 1.4$ $K = 1.4(ID)^{-0.53}$ ID from 0.3 to 4 in long radius $K = 0.75$ $K = 0.75(ID)^{-0.81}$ ID from 0.3 to 4 in	regular $K = 0.31$ $K = 0.44(ID)^{-0.23}$ ID from 1 to 25 in long radius $K = 0.22$ $K = 0.51(ID)^{-0.58}$ ID from 1 to 23 in
 45° Elbow	regular $K = 0.35$ $K = 0.35(ID)^{-0.14}$ ID from 0.3 to 4 in	long radius $K = 0.17$ $K = 0.22(ID)^{-0.14}$ ID from 1 to 23 in

Figure 8.15

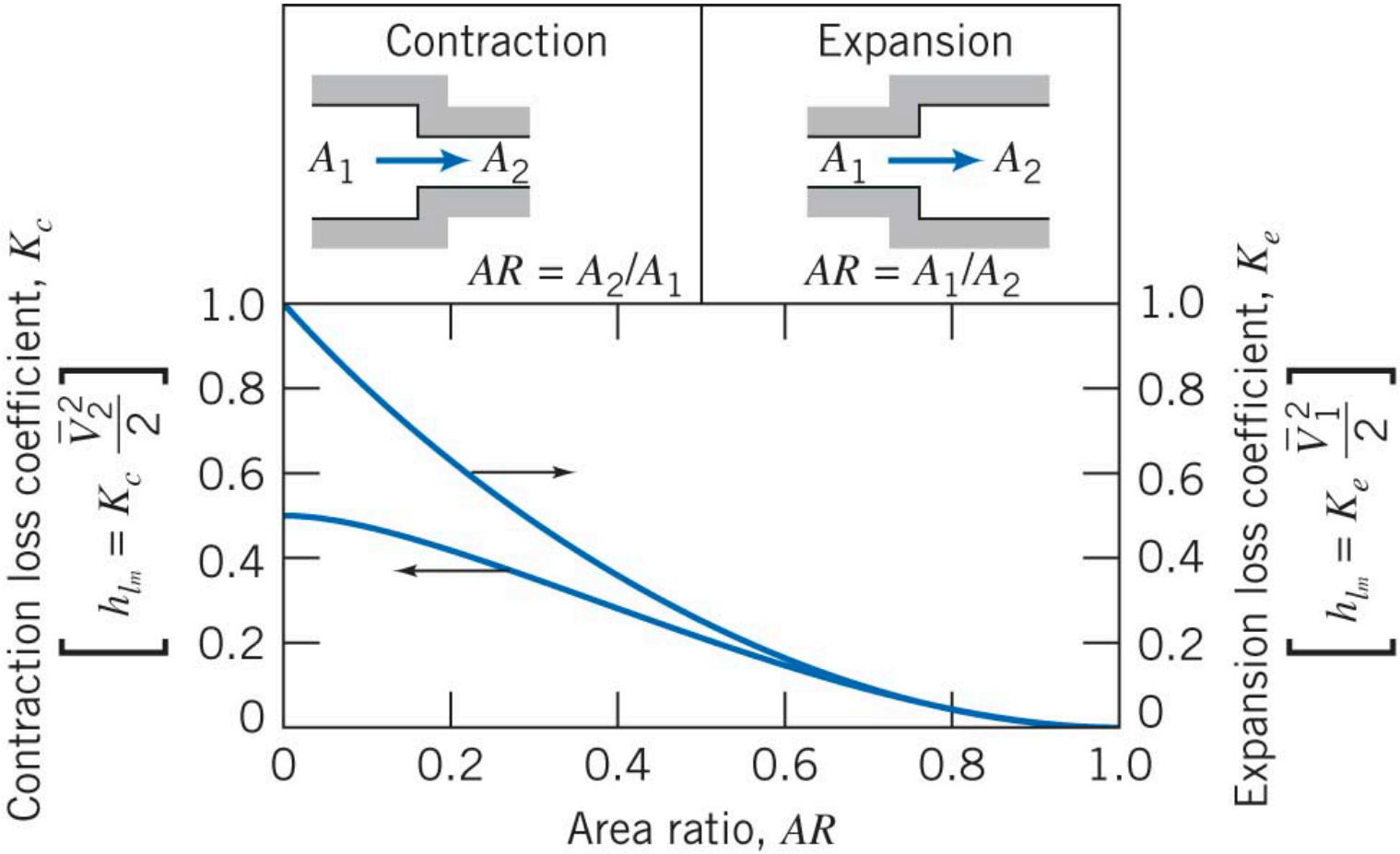
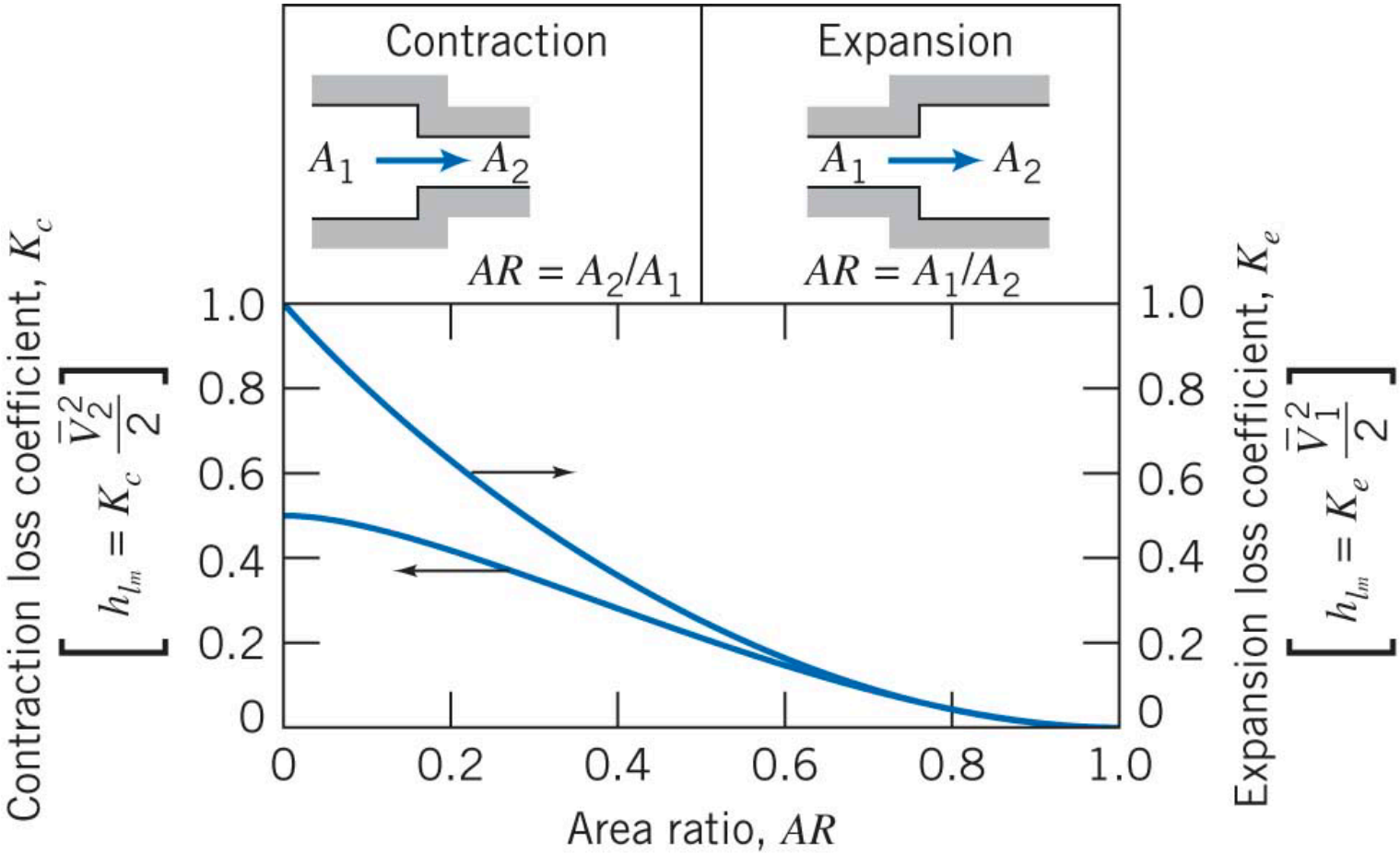

Contraction
Expansion
A_1
A_2
A_1
A_2
$AR = A_2/A_1$
$AR = A_1/A_2$
1.0
0.8
0.6
0.4
0.2
0
0
0.2
0.4
0.6
0.8
1.0
Area ratio, AR
Contraction loss coefficient, K_c
$\left[h_{l_m} = K_c \frac{\bar{V}_2^2}{2}\right]$
Expansion loss coefficient, K_e
$\left[h_{l_m} = K_e \frac{\bar{V}_1^2}{2}\right]$

Figure 8.15



Both loss coefficients are based on the larger value of velocity (i.e. the smaller area) !

Table 8.4

Representative Loss Coefficients for Fittings and Valves

Fitting	Geometry	K	Fitting	Geometry	K
90° elbow	Flanged regular	0.3	Globe valve	Open	10
	Flanged long radius	0.2	Angle valve	Open	5
	Threaded regular	1.5	Gate valve	Open	0.20
	Threaded long radius	0.7		75% open	1.10
	Miter	1.30		50% open	3.6
	Miter with vanes	0.20		25% open	28.8
45° Elbow	Threaded regular	0.4	Ball valve	Open	0.5
	Flanged long radius	0.2		1/3 closed	5.5
Tee, dividing	Threaded	0.9		2/3 closed	200
line flow	Flanged	0.2	Water meter		7
Tee, branching	Threaded	2.0	Coupling		0.08
flow	Flanged	1.0			

Source: Data from References [12] and [34].

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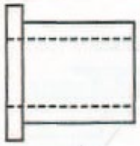
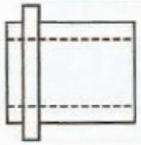
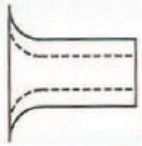
	Flanged long radius	0.2
Tee, dividing line flow	Threaded	0.9
	Flanged	0.2
Tee, branching flow	Threaded	2.0
	Flanged	1.0

Source: Data from References [12] and [34].

Loss coefficients depend on both the geometry of the bend (i.e. regular vs long radius) and the type of connection (i.e. flanged vs threaded)

Tables of Loss Coefficients - Janna textbook (3rd edition)

 Square edged inlet $K = 0.5$	Basket strainer $K = 1.3$
 Re-entrant inlet or inward projecting pipe $K = 1.0$	 Well rounded inlet or a bell mouth inlet $K = 0.05$
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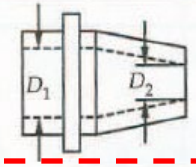 Convergent outlet or nozzle $K = 0.1(1 - D_2/D_1)$ D_2/D_1 from 0.5 to 0.9	

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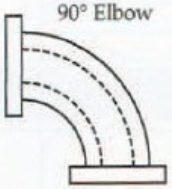
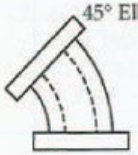
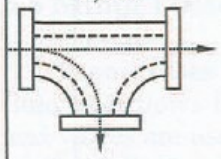
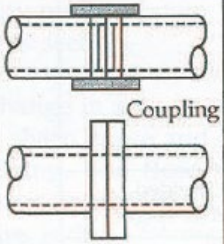
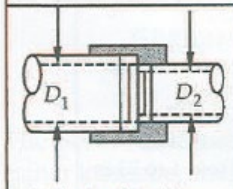
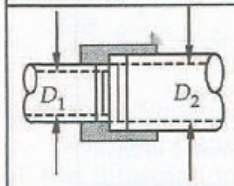
 90° Elbow	regular $K = 1.4$ $K = 1.4(ID)^{-0.53}$ ID from 0.3 to 4 in long radius $K = 0.75$ $K = 0.75(ID)^{-0.81}$ ID from 0.3 to 4 in	regular $K = 0.31$ $K = 0.44(ID)^{-0.23}$ ID from 1 to 25 in long radius $K = 0.22$ $K = 0.51(ID)^{-0.58}$ ID from 1 to 23 in
 45° Elbow	regular $K = 0.35$ $K = 0.35(ID)^{-0.14}$ ID from 0.3 to 4 in	long radius $K = 0.17$ $K = 0.22(ID)^{-0.14}$ ID from 1 to 23 in

Tables of Loss Coefficients - Janna textbook (3rd edition)

	threaded	flanged, welded, glued, bell & spigot
Return bend	regular $K = 1.5$ $K = 1.5(ID)^{-0.57}$ ID from 0.3 to 4 in	regular $K = 0.3$ $K = 0.43(ID)^{-0.26}$ ID from 1 to 23 in long radius $K = 0.2$ $K = 0.43(ID)^{-0.53}$ ID from 1 to 23 in
T joint 	line flow $K = 0.9$ all sizes ID from 0.3 to 4 in branch flow $K = 1.9$ $K = 1.9(ID)^{-0.38}$ ID from 0.3 to 4 in	line flow $K = 0.14$ $K = 0.27(ID)^{-0.46}$ ID from 1 to 20 in branch flow $K = 0.69$ $K = 1.0(ID)^{-0.29}$ ID from 1 to 20 in
Coupling 	$K = 0.08$ $K = 0.083(ID)^{-0.69}$ ID from 0.4 to 4 in	$K = 0.08$ ID from 0.3 to 23 in
Reducing bushing 	$K = 0.5 - 0.167(D_2/D_1) - 0.125(D_2/D_1)^2 - 0.208(D_2/D_1)^3$ $0.25 < D_2/D_1 < 1$	
Sudden expansion 	$K = ((D_2/D_1)^2 - 1)^2$ $1 < D_2/D_1 < 5$	

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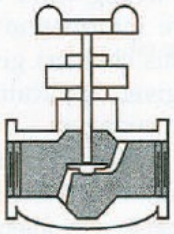
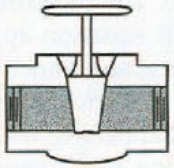
Tee, dividing	Threaded	0.9		2/3 closed	200
line flow	Flanged	0.2	Water meter		7
Tee, branching	Threaded	2.0	Coupling		0.08
flow	Flanged	1.0			

Source: Data from References [12] and [34].

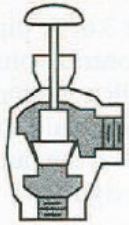
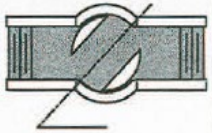
Depending on the exact conditions “minor losses” can be a significant source of energy loss!!

Tables of Loss Coefficients - Janna textbook (3rd edition)

The amount of resistance to flow when in the 'fully open' position varies with their design

	threaded	flanged, welded, glued, bell & spigot
Globe valve 	fully open $K = 10$ $K = \exp\{2.158 - 0.459 \ln(ID) + 0.259[\ln(ID)]^2 - 0.123[\ln(ID)]^3\}$ ID from 0.3 to 4 in	fully open $K = 10$ $K = \exp\{2.565 - 0.916 \ln(ID) + 0.339[\ln(ID)]^2 - 0.01416[\ln(ID)]^3\}$ ID from 0.3 to 4 in
Gate Valve 	fully open $K = 0.15$ $K = 0.24(ID)^{-0.47}$ ID from 0.3 to 4 in	fully open $K = 0.15$ $K = 0.78(ID)^{-1.14}$ ID from 1 to 20 in
	All sizes	

Fraction closed	0	1/4	3/8	1/2	5/8	3/4	7/8
K =	0.15	0.26	0.81	2.06	5.52	17.0	97.8

 Angle Valve	fully open $K = 2.0$ $K = 4.5(ID)^{-1.08}$ ID from 0.6 to 4 in	fully open $K = 2.0$ $K = \exp\{1.569 - 1.43 \ln(ID) + 0.8[\ln(ID)]^2 - 0.137[\ln(ID)]^3\}$ ID from 1 to 20 in
 Ball Valve	All sizes	

°	0	10	20	30	40	50	60	70	80
K =	0.05	0.29	1.56	5.47	17.3	25.6	206	485	∞

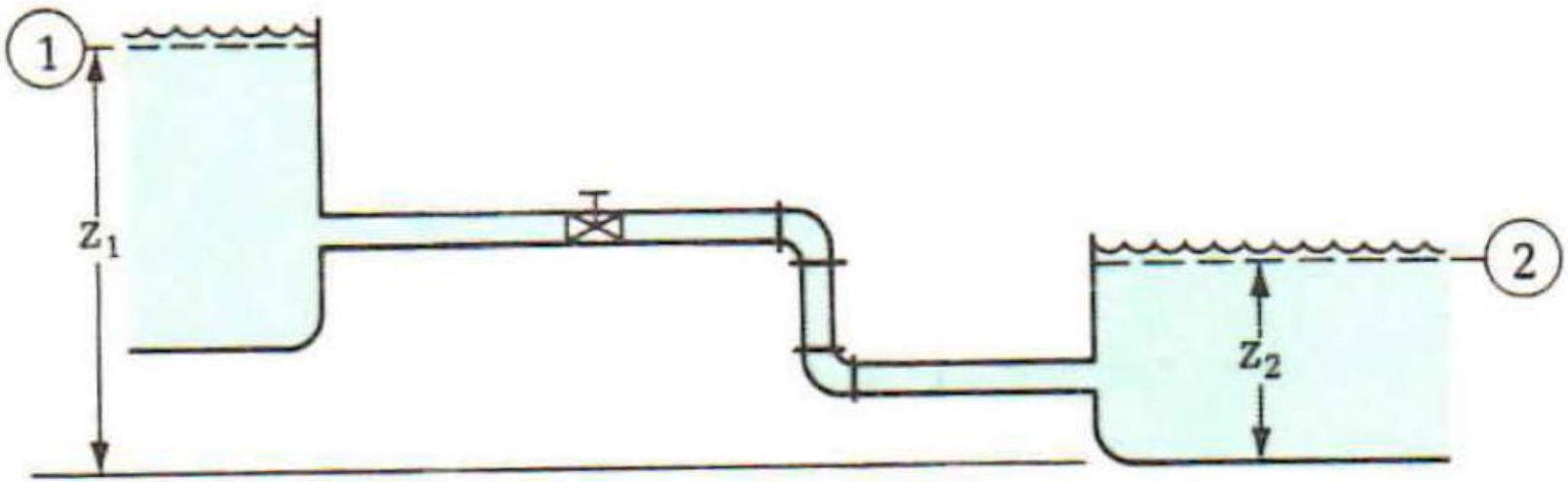
Check Valves		
Swing Type	$K = 2.5$	$K = 2.5$
Ball Type	$K = 70.0$	$K = 70.0$
Lift Type	$K = 12.0$	$K = 12.0$

https://www.nuclear-power.net/wp-content/uploads/2016/06/table-of-K-values-valves-elbows-bends.png

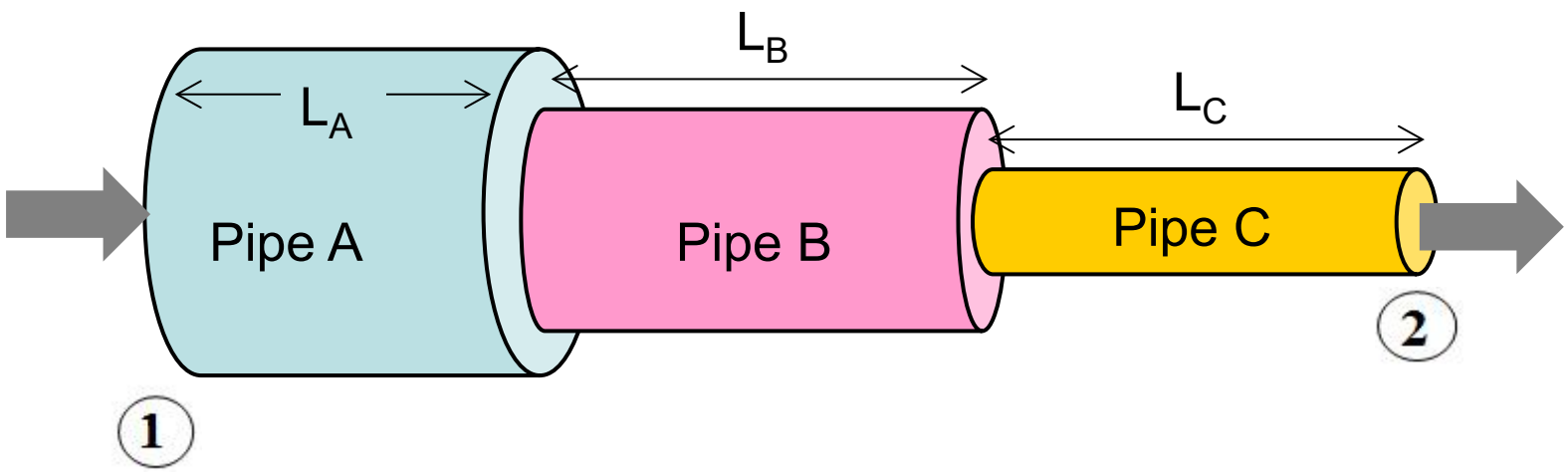
Fitting	Type	K
Gate valve	Fully Open	0.17
	3/4 Open	0.9
	1/2 Open	4.5
	1/4 Open	24
Plug disk	Fully Open	9
	3/4 Open	13
	1/2 Open	36
	1/4 Open	112
Butterfly valve	$\theta = 5°$	0.24
	$\theta = 10°$	0.52
	$\theta = 20°$	1.54
	$\theta = 40°$	10.8
	$\theta = 60°$	118

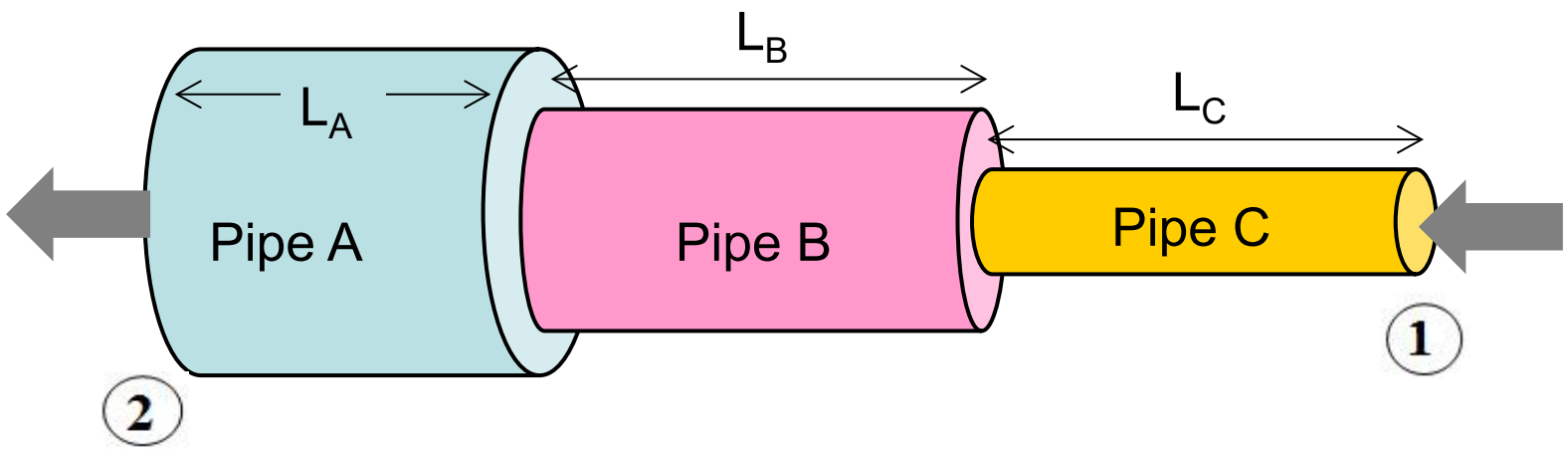
https://videos.emerson.com/detail/videos/fisher/video/5806202349001/what-are-the-components-of-a-butterfly-valve?autoStart=true

Bernoulli Equation – Minor losses (h_{lm})



$$h_{lm} = \sum \left[K \frac{V_c^2}{2} \right] =$$


L_B
L_A
L_C
Pipe A
Pipe B
Pipe C
2
1


L_B
L_A
L_C
Pipe A
Pipe B
Pipe C
1
2

Single-Path Systems

Type I → Determine pressure drop

Type II → Determine pipe length

Type III → Determine flow rate

Type IV → Choose the ‘best’ pipe diameter

Single-Path Systems

Type I → Determine pressure drop

Given flow rate, pipe diameter, pipe length, and fluid properties

Tabulate data for minor loss coefficients

↓

Calculate relative roughness (e/D)

↓

Calculate Reynolds number (Re)

↓

Determine friction factor (f)

↓

Calculate pressure drop (ΔP) using Bernoulli Equation

Review Example 8.5 from 10th (& 9th) edition textbook

Single-Path Systems

Type II → Determine pipe length

Given pressure drop, flow rate, pipe diameter, and fluid properties

Tabulate data for minor loss coefficients

↓

Calculate relative roughness (e/D)

↓

Calculate Reynolds number (Re)

↓

Determine friction factor (f)

↓

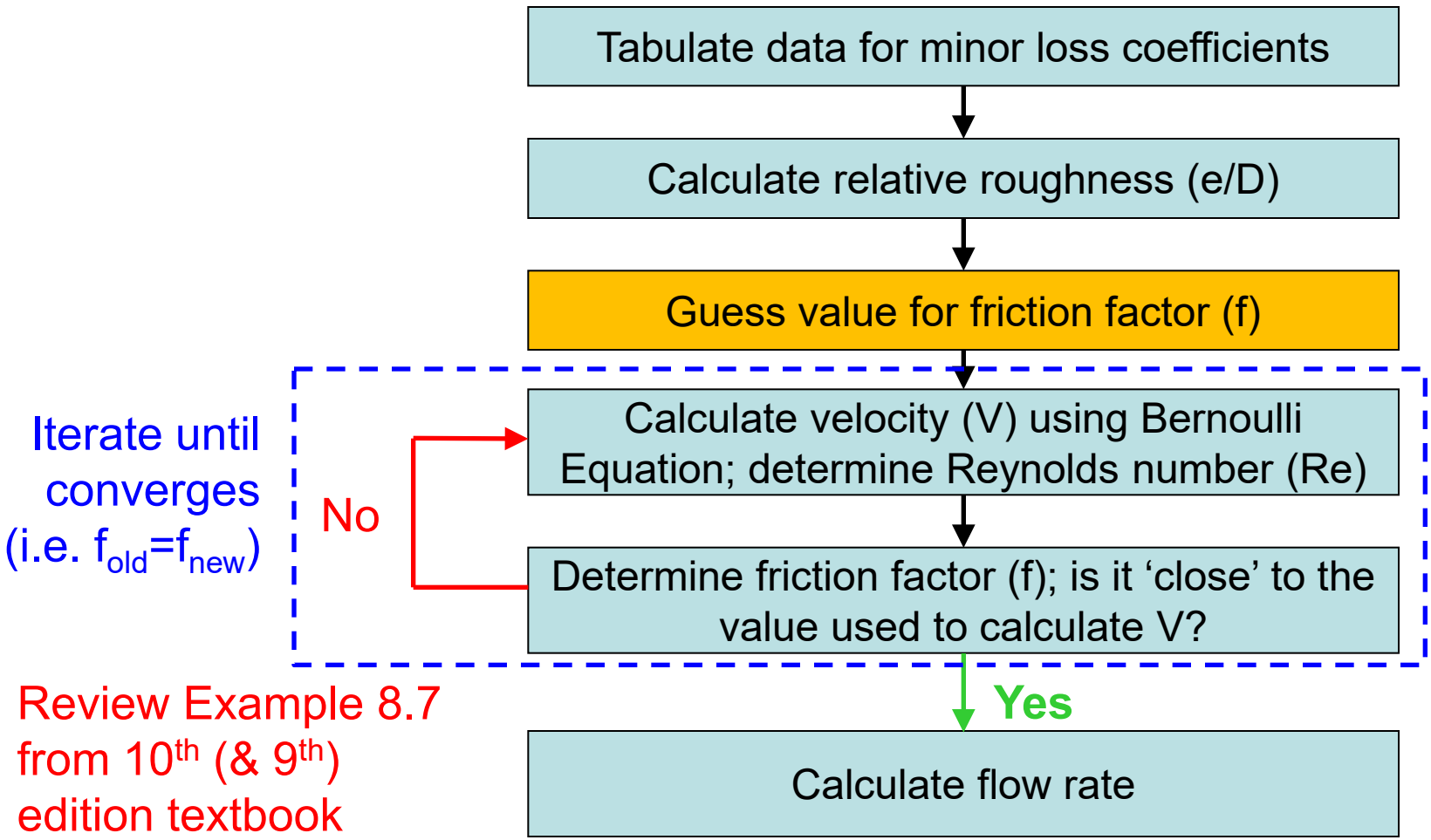
Calculate pipe length (L) using Bernoulli Equation

Review Example 8.6 from 10th (& 9th) edition textbook

Single-Path Systems

Type III → Determine flow rate

Given pressure drop, pipe dimensions, and fluid properties



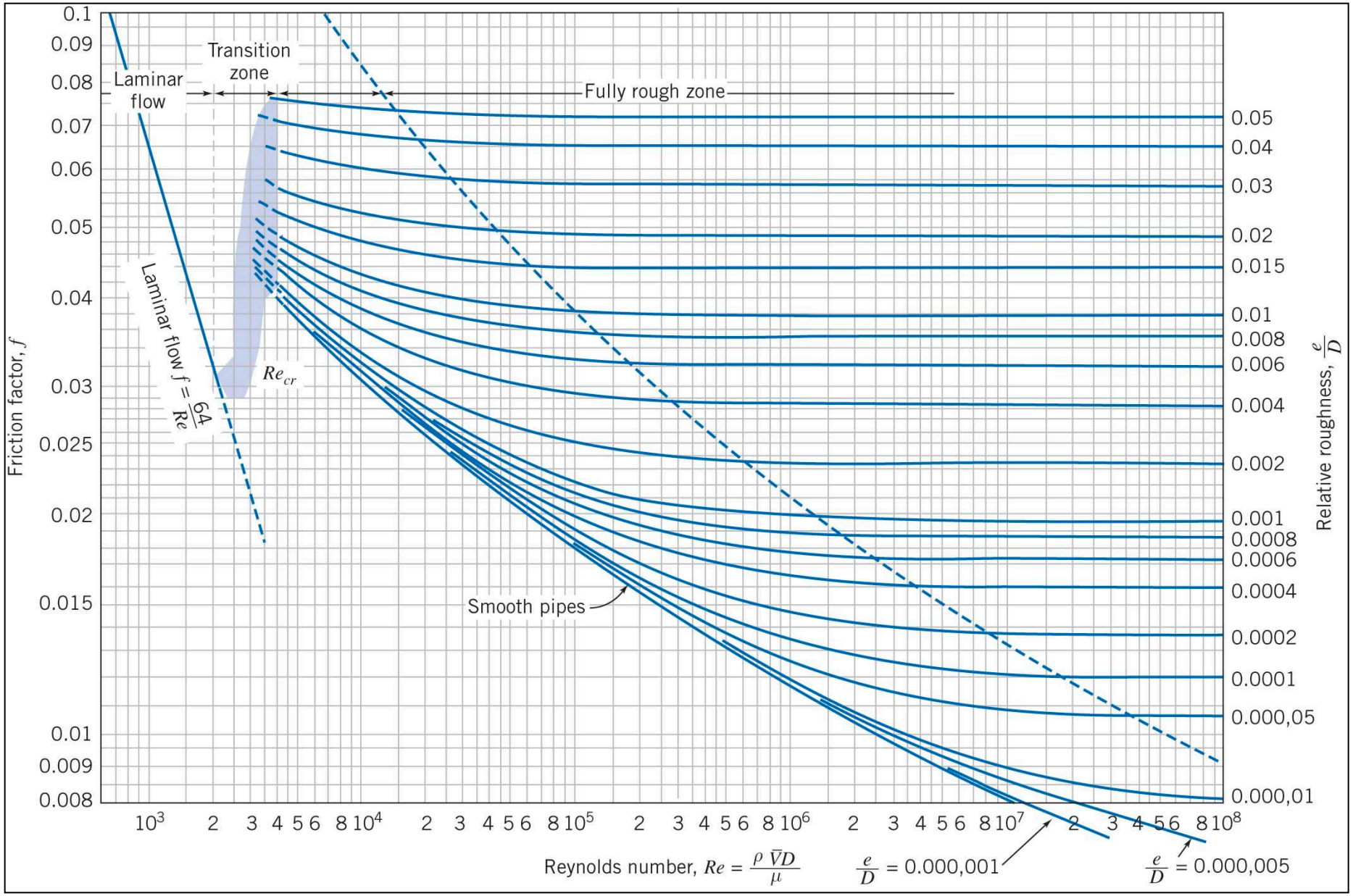
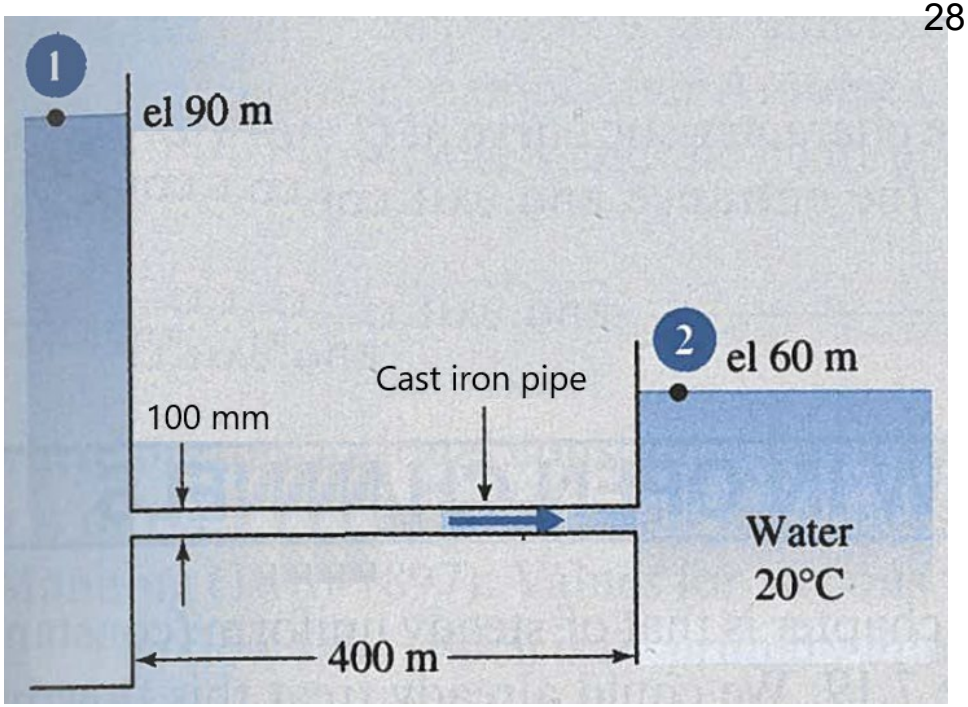

Laminar flow
Transition zone
Fully rough zone
Laminar flow $f = \frac{64}{Re}$
Re_{cr}
Smooth pipes
Friction factor, f
Relative roughness, $\frac{e}{D}$
Reynolds number, $Re = \frac{\rho \bar{V} D}{\mu}$
$\frac{e}{D}$ = 0.000,001
$\frac{e}{D}$ = 0.000,005
0.1
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.025
0.02
0.015
0.01
0.009
0.008
10^3 2 3 4 5 6 8 10^4 2 3 4 5 6 8 10^5 2 3 4 5 6 8 10^6 2 3 4 5 6 8 10^7 2 3 4 5 6 8 10^8
0.05
0.04
0.03
0.02
0.015
0.01
0.008
0.006
0.004
0.002
0.001
0.0008
0.0006
0.0004
0.0002
0.0001
0.000,05
0.000,01

Figure 8.13

Determine the volumetric flow rate of water in the cast iron pipe (internal diameter = 100 mm) between the two tanks as shown in the figure.



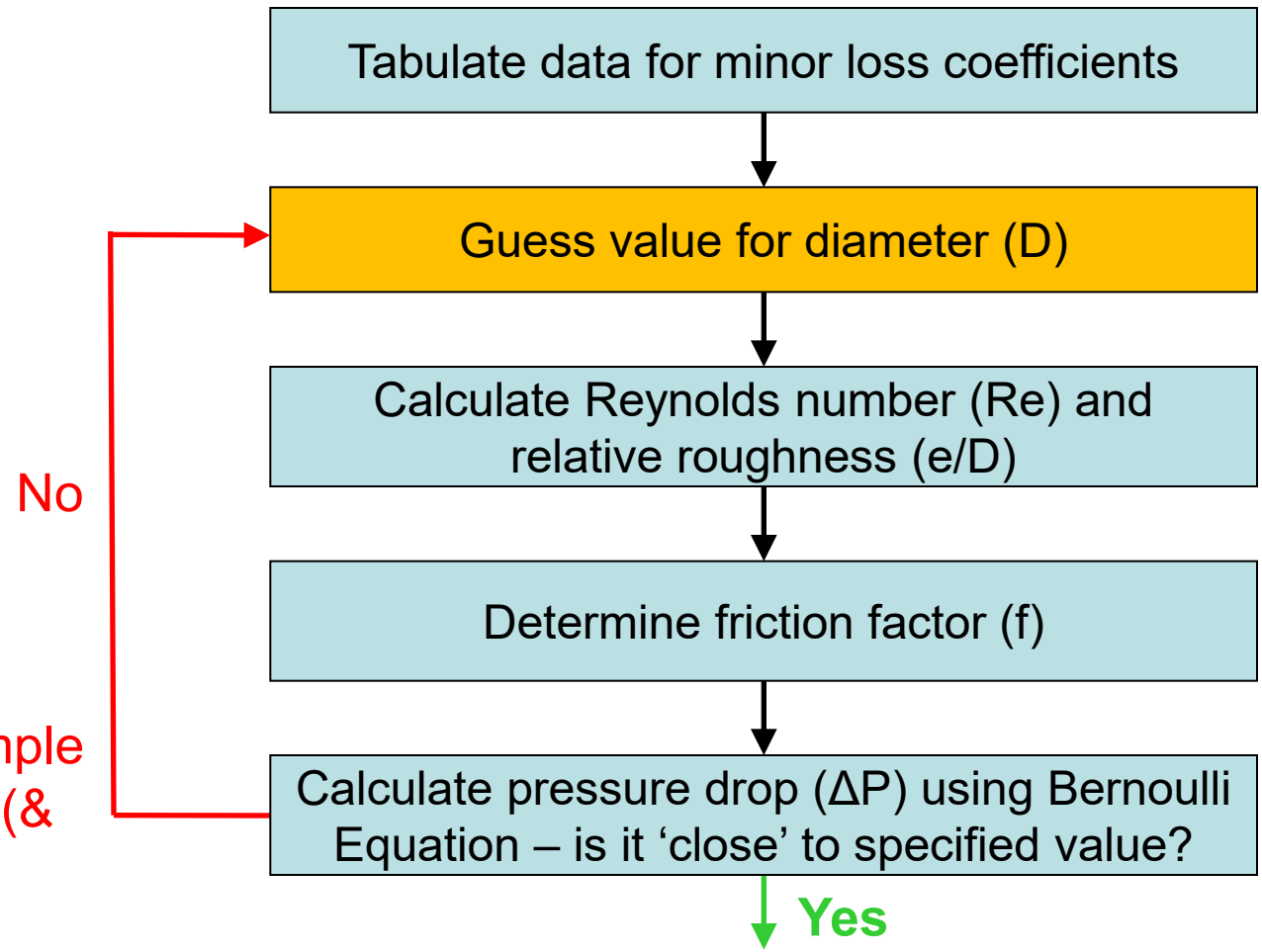
(Answer = 0.0186 m^3/s)

Single-Path Systems

"the minimum diameter (for minimum pipe cost) that will deliver the design flow rate"

Type IV → Choose the 'best' pipe diameter

Given pressure drop, flow rate, and fluid properties



Review Example 8.8 from 10th (& 9th) edition textbook

Southland®

PIPE • NIPPLES • FITTINGS

A BRAND OF MUELLER INDUSTRIES

BLACK THREADED STEEL PIPE			
Diameter	Length	Part No.	Per Piece
1/8 ***	10	580-1200HC	4.71
1/4 ***	10	581-1200HC	6.83
3/8 ***	10	582-1200HC	8.37
1/2	10	583-1200HC	8.65
3/4	10	584-1200HC	11.22
1	10	585-1200HC	16.20
1-1/4	10	586-1200HC	21.81
1-1/2	10	587-1200HC	26.04
2	10	588-1200HC	35.02
2-1/2	10	589-1200HC	57.20
3	10	590-1200HC	74.87
4	10	591-1200HC	105.50

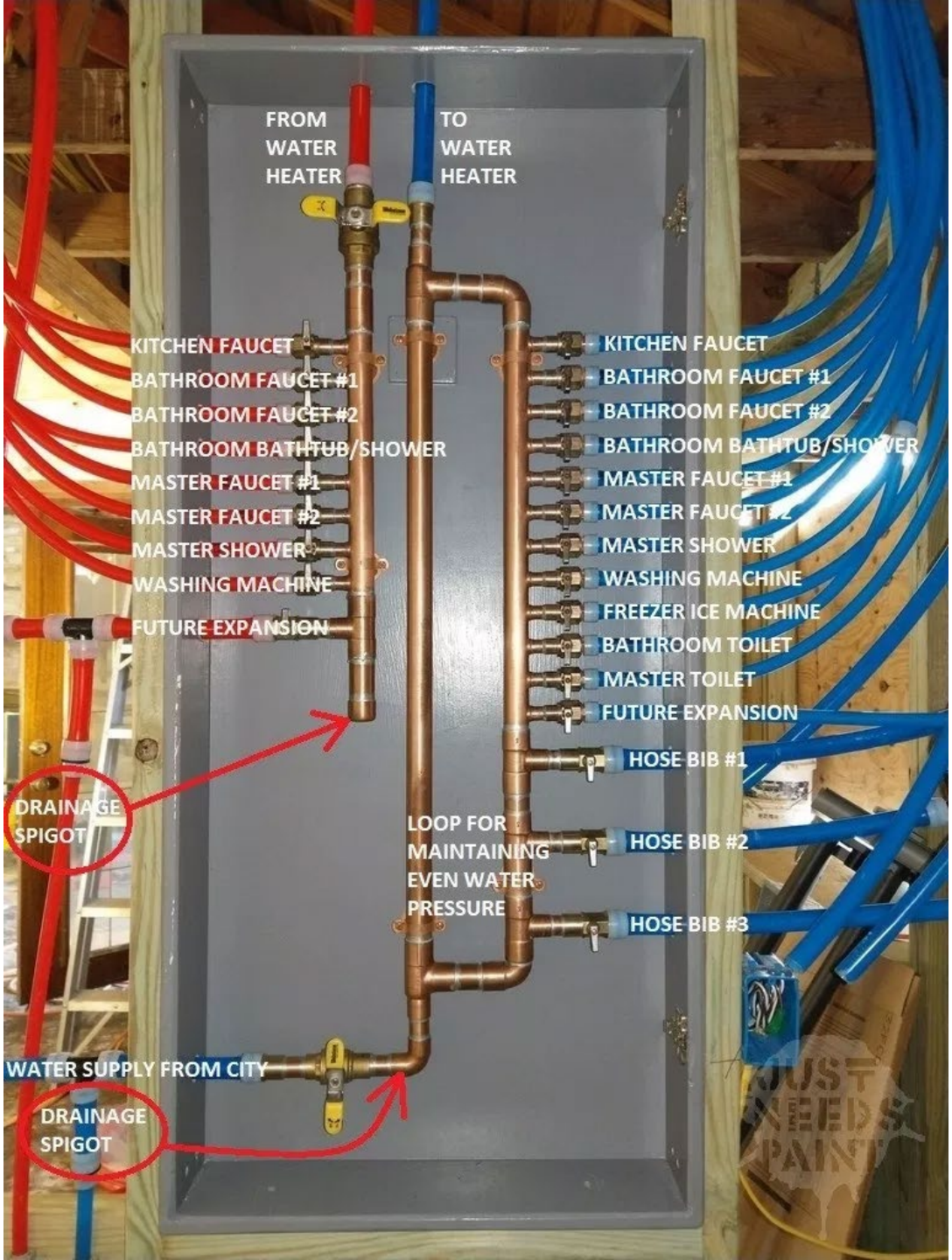

FROM WATER HEATER
TO WATER HEATER
KITCHEN FAUCET
BATHROOM FAUCET #1
BATHROOM FAUCET #2
BATHROOM BATHTUB/SHOWER
MASTER FAUCET #1
MASTER FAUCET #2
MASTER SHOWER
WASHING MACHINE
FUTURE EXPANSION
KITCHEN FAUCET
BATHROOM FAUCET #1
BATHROOM FAUCET #2
BATHROOM BATHTUB/SHOWER
MASTER FAUCET #1
MASTER FAUCET #2
MASTER SHOWER
WASHING MACHINE
FREEZER ICE MACHINE
BATHROOM TOILET
MASTER TOILET
FUTURE EXPANSION
HOSE BIB #1
HOSE BIB #2
HOSE BIB #3
DRAINAGE SPIGOT
LOOP FOR MAINTAINING EVEN WATER PRESSURE
WATER SUPPLY FROM CITY
DRAINAGE SPIGOT

Table 8.5

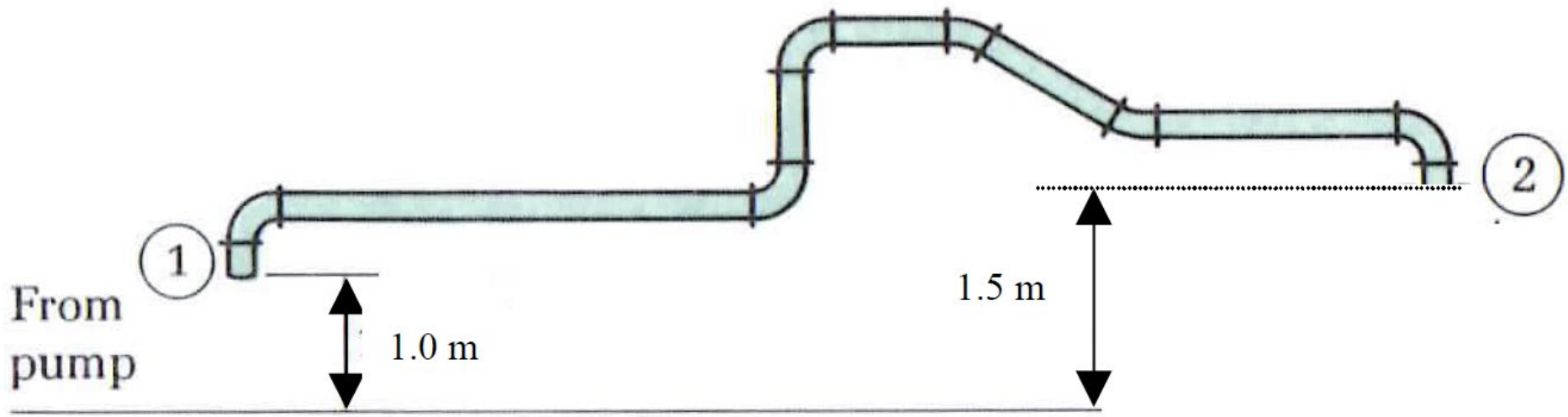
Standard Sizes for Carbon Steel, Alloy Steel, and Stainless Steel Pipe

Nominal Pipe Size (in.)	Inside Diameter (in.)	Nominal Pipe Size (in.)	Inside Diameter (in.)
$\frac{1}{8}$	0.269	$2\frac{1}{2}$	2.469
$\frac{1}{4}$	0.364	3	3.068
$\frac{3}{8}$	0.493	4	4.026
$\frac{1}{2}$	0.622	5	5.047
$\frac{3}{4}$	0.824	6	6.065
1	1.049	8	7.981
$1\frac{1}{2}$	1.610	10	10.020
2	2.067	12	12.000

Source: Data from Reference [11].

See additional table of pipe specifications (from Janna textbook) posted on AVENUE

A flow rate of 0.2 m^3/min of carbon tetrachloride is required in a processing plant. The total piping system show below is 50 m in length and is to be constructed from schedule 40 galvanized threaded steel pipe (e = 0.15 mm) with threaded fittings.
If the existing pump can supply the required flow rate if the pressure drop in the piping system is less than 150 kPa, select the 'best' pipe size to use for the installation.



1st Law of Thermodynamics

- By conservation of mass: $\dot{m} = \rho_1 V_1 A_1 = \rho_2 V_2 A_2$

$$0 = \dot{m}\left[\left(\frac{P_2}{\rho_2} + \alpha_2 \frac{V_2^2}{2} + gz_2\right) - \left(\frac{P_1}{\rho_1} + \alpha_1 \frac{V_1^2}{2} + gz_1\right)\right] + \dot{m}(u_2 - u_1) - \dot{Q} + \dot{W}_{shaft}$$

- ***Assumption #3*** → **No shaft work (i.e. pumps, turbines) occurs**
- *Assumption #4* → Incompressible fluid ($\rho_1 = \rho_2 = \rho$)

$$0 = \left(\frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2\right) - \left(\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1\right) + \left(u_2 - u_1 - \frac{\dot{Q}}{\dot{m}}\right) \qquad \text{Eqn 6.14}$$

- *Assumption #5* → Frictionless flow; $u_2 - u_1 - \frac{\dot{Q}}{\dot{m}} = 0$

$$0 = \left(\frac{P_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + gz_2\right) - \left(\frac{P_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + gz_1\right)$$

Single-Path Systems with a Pump

PUMPS

Numerous types → more details in next section of course

For now:

- Adds energy to the fluid in the form of a gain in pressure
- Pump Head:

$$H_{pump} = \frac{\Delta P_{pump}}{\rho g}$$

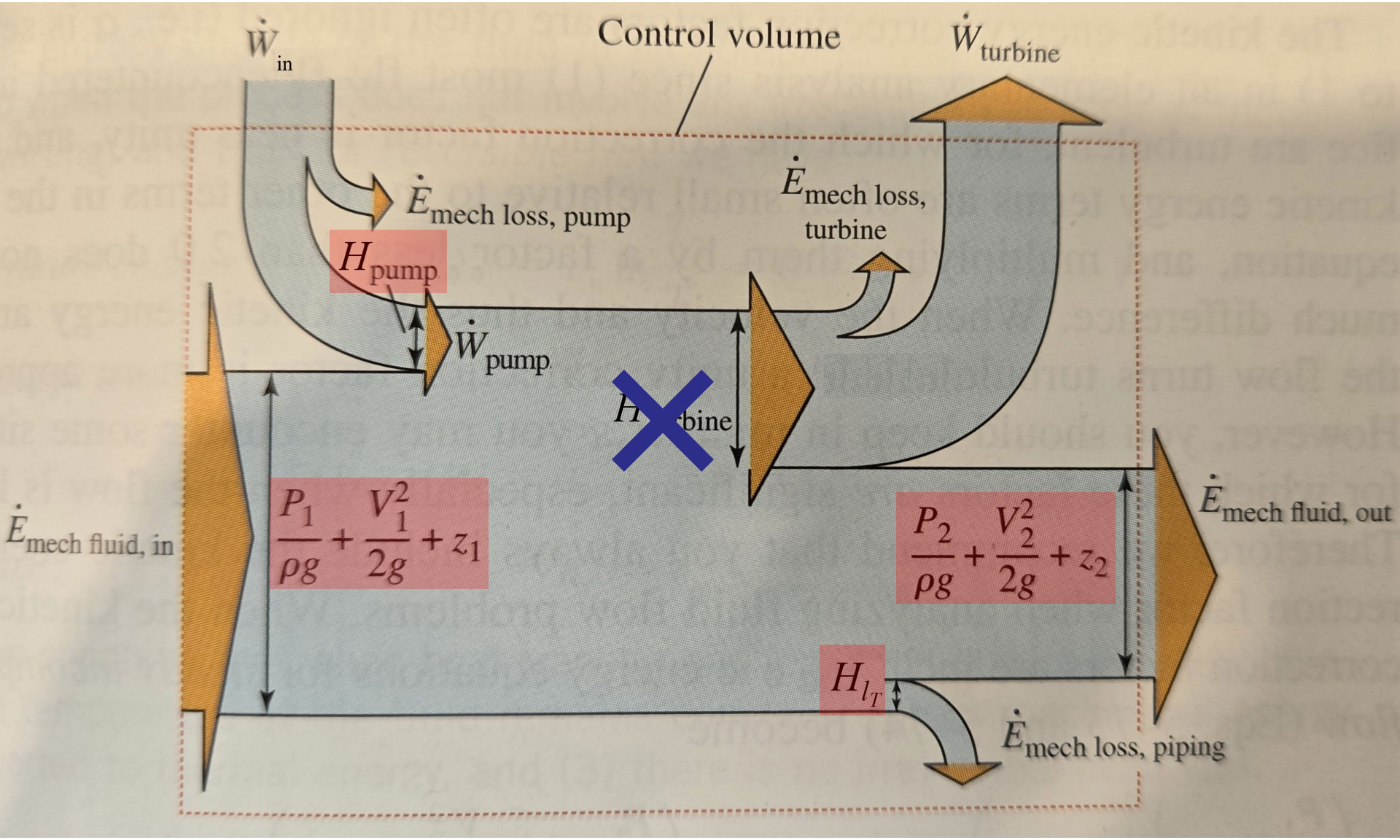
Eqn 8.46

- The head (i.e. energy) required to be supplied by the pump increases with increasing flow rate

For the Bernoulli Equation:

$$\frac{P_1}{\rho g} + \alpha_1 \frac{V_1^2}{2g} + z_1 + H_{pump} = \frac{P_2}{\rho g} + \alpha_2 \frac{V_2^2}{2g} + z_2 + H_l + H_{l_m}$$

Sankey diagram of the full Bernoulli Equation



Characterizing the pump performance

$$h_{pump} = \frac{\Delta P_{pump}}{\rho}$$ Eqn 8.46

$$\underbrace{\dot{m} \cdot h_{pump}} = \dot{m} \cdot \frac{\Delta P_{pump}}{\rho}$$

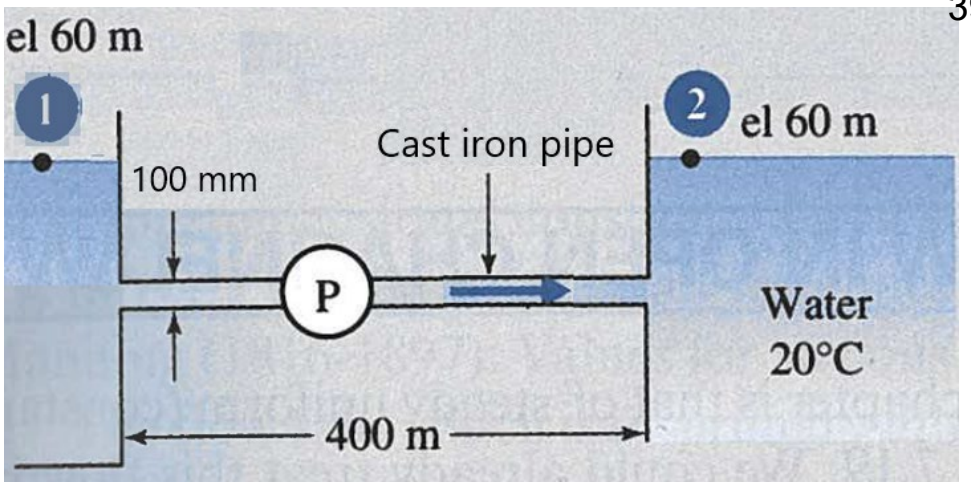
$$\dot{W}_{pump} = \frac{\dot{m}}{\rho} \Delta P_{pump} = Q \cdot \Delta P_{pump}$$ Eqn 8.47

"Power reaching the fluid"

Losses in a machine, like a pump, mean that the *power input to the pump* $(\dot{W}_{input})$ is ALWAYS larger than the *power that reaches the fluid*!

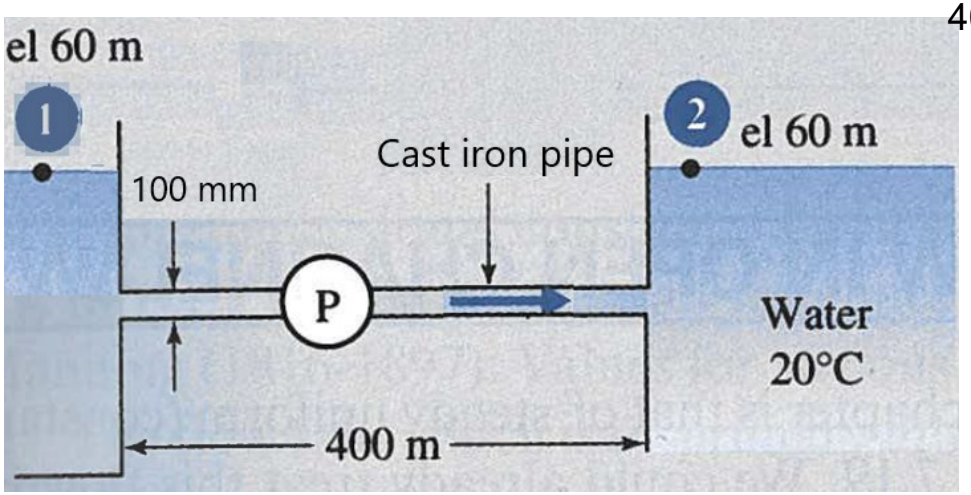
Define Pump Efficiency: $$\eta_P \equiv \frac{\dot{W}_{pump}}{\dot{W}_{input}} 100\%$$ Eqn 8.48

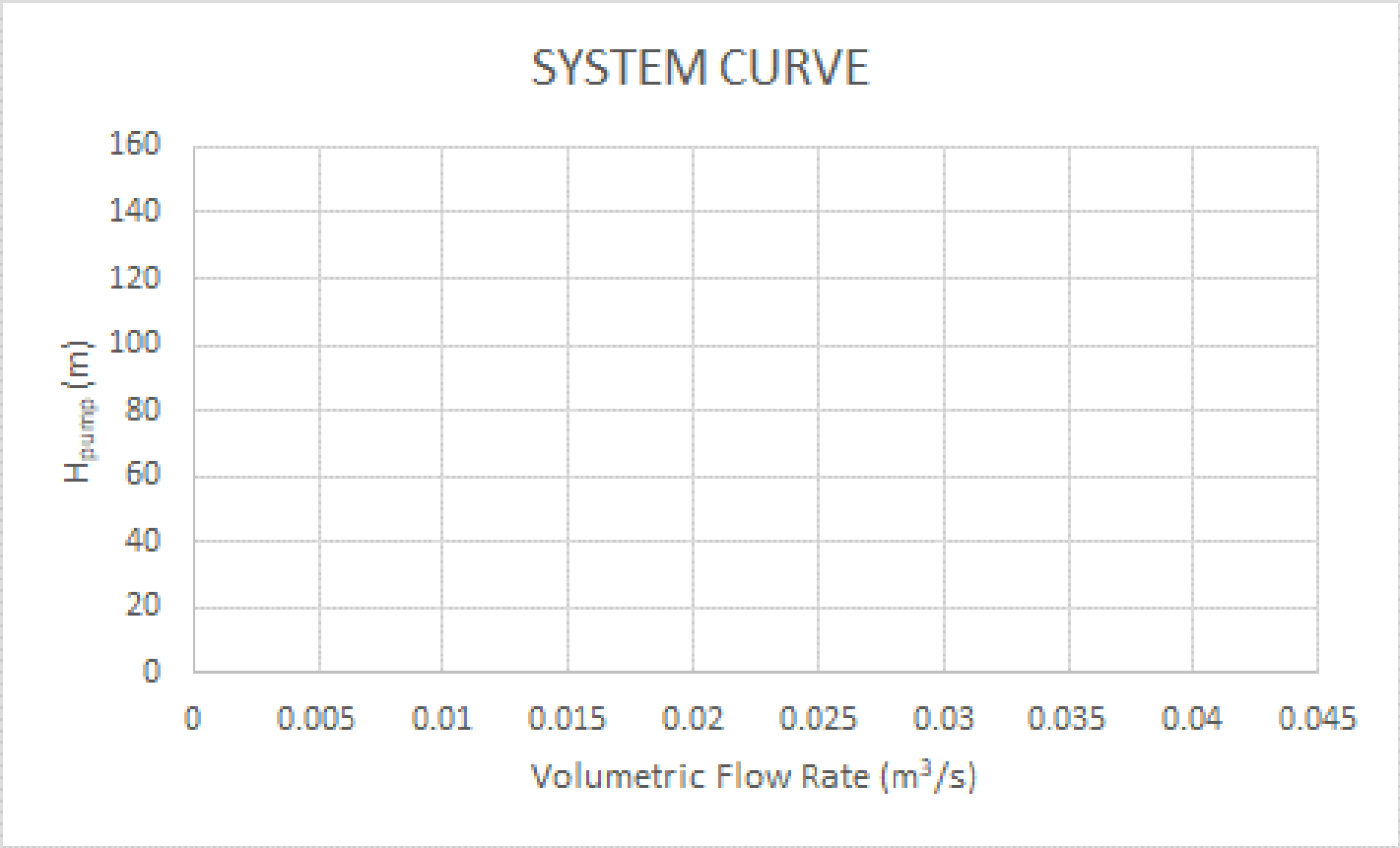
For the different values of H_{pump} in the table below, determine the volumetric flow rate of water in the cast iron pipe (internal diameter = 100 mm) between the two tanks as shown in the figure.



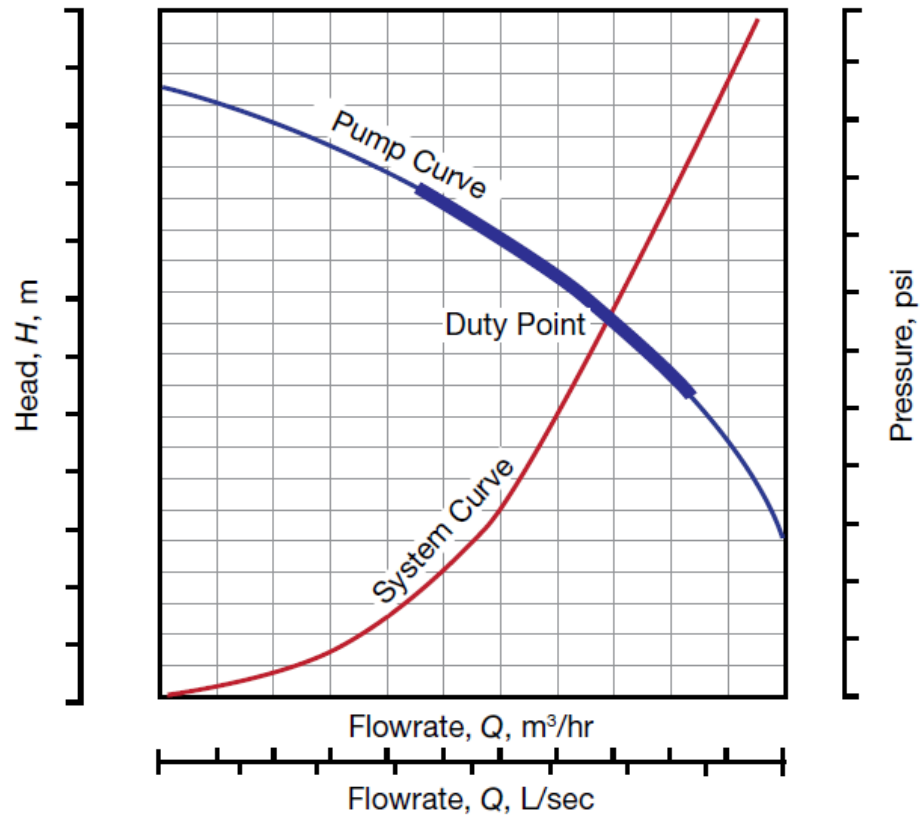
H_{pump} (m)	Q (m^3/s)
0	
10	
35	
80	
140	

For the different values of H_{pump} in the table below, determine the volumetric flow rate of water in the cast iron pipe (internal diameter = 100 mm) between the two tanks as shown in the figure.





Pump Sizing: Bridging the Gap Between Theory and Practice



▲ **Figure 4.** A basic pump curve plots pressure (or head) as a function of flowrate.

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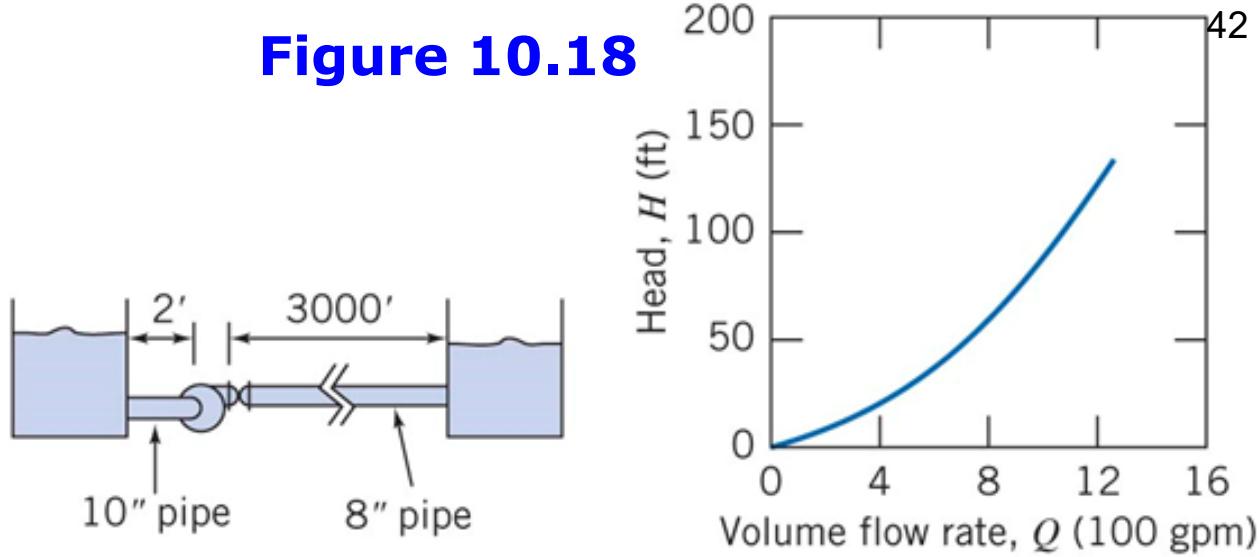
Review Example 10.8 from 10th (& 9th) edition textbook

Figure 10.18



(a) Purely frictional pressure drop

Simplified form of Bernoulli Eqn → $H_{pump} = H_l + H_{l_m}$

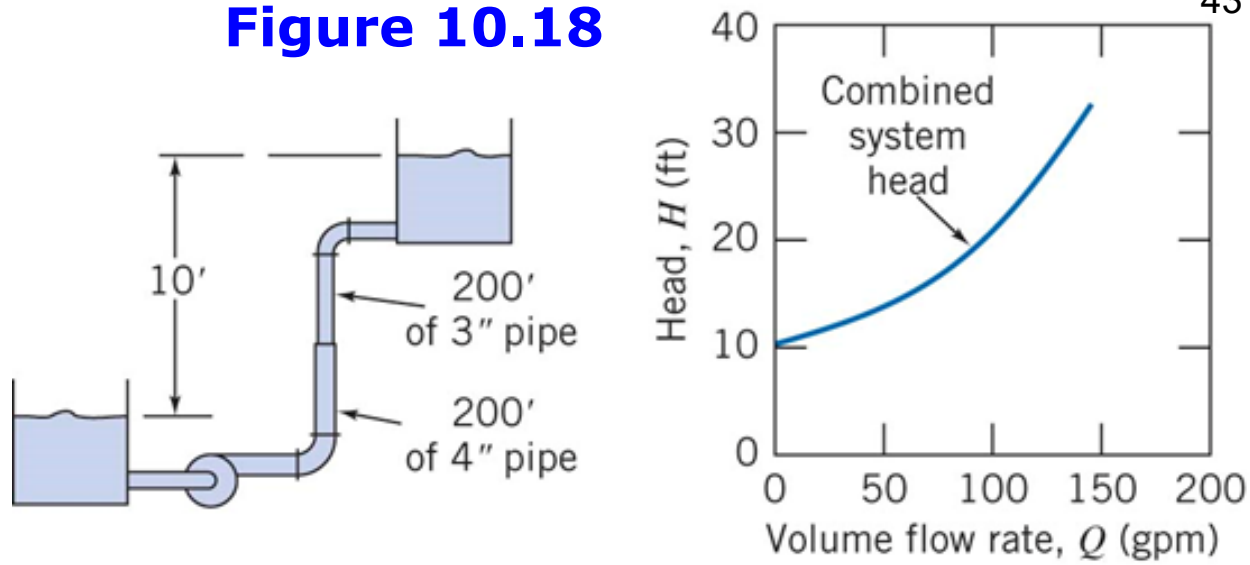
$$H_{pump} = \left[f\frac{L}{D_h}\frac{V^2}{2g}\right]_{10"pipe} + \left[f\frac{L}{D_h}\frac{V^2}{2g}\right]_{8"pipe} + \sum\left[K\frac{V_c^2}{2g}\right]$$

$$H_{pump} = \left[f\frac{L}{D_h}\frac{\left(Q/A\right)^2}{2g}\right]_{10"pipe} + \left[f\frac{L}{D_h}\frac{\left(Q/A\right)^2}{2g}\right]_{8"pipe} + \sum\left[K\frac{\left(Q/A\right)^2}{2g}\right]$$

$$H_{pump} = \left\{\left[f\frac{L}{D_h}\frac{1}{2gA^2}\right]_{10"pipe} + \left[f\frac{L}{D_h}\frac{1}{2gA^2}\right]_{8"pipe} + \sum\left[K\frac{1}{2gA^2}\right]\right\}Q^2$$

Review Example 10.8 from 10th (& 9th) edition textbook

Figure 10.18



(b) Combination of frictional and gravity pressure changes

Simplified form of Bernoulli Eqn →

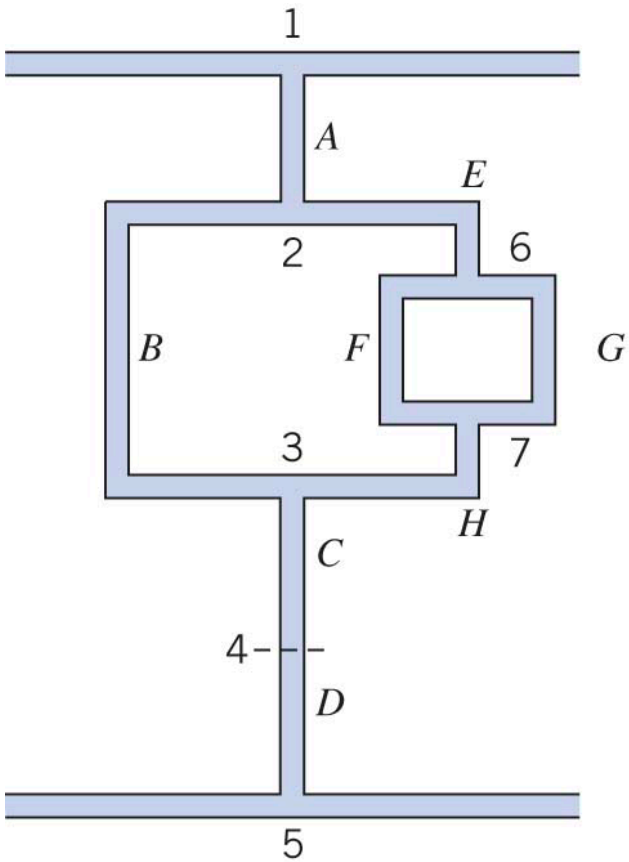
$$z_1 + H_{pump} = z_2 + H_l + H_{l_m}$$

$$H_{pump} = z_2 - z_1 + \left[f\frac{L}{D_h}\frac{V^2}{2g}\right]_{10"pipe} + \left[f\frac{L}{D_h}\frac{V^2}{2g}\right]_{8"pipe} + \sum\left[K\frac{V_c^2}{2g}\right]$$

$$H_{pump} = z_2 - z_1 + \left\{\left[f\frac{L}{D_h}\frac{1}{2gA^2}\right]_{10"pipe} + \left[f\frac{L}{D_h}\frac{1}{2gA^2}\right]_{8"pipe} + \sum\left[K\frac{1}{2gA^2}\right]\right\}Q^2$$

Multi-Path Systems

Real-world pipe systems consist of network of pipes in complicated configuration; for example;



A: L = 10 ft,	D = 1.5 in.
B: L = 20 ft,	D = 1.5 in.
C: L = 10 ft,	D = 2 in.
D: L = 10 ft,	D = 1.5 in.
E: L = 5 ft,	D = 1.5 in.
F: L = 10 ft,	D = 1 in.
G: L = 10 ft,	D = 1.5 in.
H: L = 5 ft,	D = 2 in.

Figure 8.17 – Schematic of part of a pipe network

Multi-Path Systems

- **Manual solutions do exist (e.g. Hardy-Cross method)**

BUT

  - Require iterative solution methods
  - Much more difficult to solve
  - Very time consuming

- **Various 'piping systems' software packages:**
  - EPANET (freeware)
  - Simcenter Flomaster
  - OpenFlows WaterCAD
  (https://www.bentley.com/en/products/product-line/hydraulics-and-hydrology-software/watercad)
  - AFT Fathom
  (http://www.aft.com/products/fathom)
  - PIPE-FLO

End of material that will be covered on the 2nd midterm

Friday March 20th
7:00pm – 9:00pm

MDCL 1102 & 1105

Problems from textbook for self-study

10th Edition	9th Edition
8.28	8.83
8.29	8.97
8.31	8.75
8.33	8.80
8.38	8.107
8.43	8.132
8.46	8.133

PIPE-FLO

Tutorial 5 “Fluid Statics with Minor Losses”

Tutorial 6 “Other Devices, Valves, & Meters”